GHULAM ISHAQ KHAN INSTITUTE

of Engineering Sciences and Technology

Faculty of Computer Science and Engineering

CE342L • Computational Methods and Tools Lab

Lab Report #11

Householder Method for Tridiagonalization

Converting Symmetric Matrices to Tridiagonal Form in MATLAB

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The objective of this lab is to implement the **Householder Transformation Method** in MATLAB to convert a symmetric matrix into **tridiagonal form**. A tridiagonal matrix has nonzero entries only on the main diagonal and the two adjacent sub- and super-diagonals, which greatly simplifies subsequent eigenvalue computations.

By the end of this lab, students will be able to:

- Understand the Householder reflection and its geometric interpretation.
- Compute the Householder vector $\mathbf{V}$ and reflector matrix P.
- Apply successive Householder transformations to reduce a symmetric matrix to tridiagonal form.
- Extend the method from 3×3 to 4×4 and general $N \times N$ matrices in MATLAB.

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▶ $Householder Reflector$ A **Householder matrix** (reflector) is defined as:

$$P = I - 2\mathbf{V}\mathbf{V}^T, \qquad \|\mathbf{V}\| = 1$$

It is symmetric ($P^T = P$) and orthogonal ($P^T P = I$), so applying $A_1 = PAP$ preserves all eigenvalues of A.

▶ $Algorithm for a$**3**$\times$3 **Symmetric Matrix** Given symmetric A, one Householder step ($n - 2 = 1$ step) is needed.

Step 1 — Compute S:

$$S = \sqrt{a_{12}^2 + a_{13}^2}$$

Step 2 — Construct vector $\mathbf{V} = [0; x_2; x_3]$:

$$x_2 = \sqrt{\frac{1}{2}\left(1 + \frac{a_{12} \cdot \text{sign}(a_{12})}{S}\right)}, \qquad x_3 = \frac{a_{13} \cdot \text{sign}(a_{12})}{2S\, x_2}$$

Step 3 — Compute Householder matrix:

$$P = I - 2\mathbf{V}\mathbf{V}^T$$

Step 4 — Compute tridiagonal matrix:

$$A_1 = P\,A\,P$$

▶ *Extensionto*$\mathbf{4}\times 4$ **(Two Steps)** For a 4×4 matrix, $n - 2 = 2$ Householder steps are required.

Step 1 ($k = 1$): Eliminate a_{13} and a_{14}. Compute $S_1 = \sqrt{a_{21}^2 + a_{31}^2 + a_{41}^2}$, build $\mathbf{V}_1 = [0;\ x_2;\ x_3;\ x_4]$, form $P_1 = I - 2V_1V_1^T$, and obtain $A_1 = P_1AP_1$.

Step 2 ($k = 2$): Eliminate a_{24}. From A_1, compute $S_2 = \sqrt{A_1(3,2)^2 + A_1(4,2)^2}$, build $\mathbf{V}_2 = [0;\ 0;\ x_3;\ x_4]$, form $P_2 = I - 2V_2V_2^T$, and obtain $A_2 = P_2A_1P_2$.

▶ *General*$\mathbf{N}\times N$ **Algorithm** For an $N \times N$ symmetric matrix, $n - 2$ Householder transformations are applied. At step k ($k = 1, \ldots, n - 2$), the reflector P_k is built from a vector $\mathbf{V}_k$ that is zero in its first k entries and acts on the sub-column below position $(k + 1, k)$.

▶ *BaseMATLABCode(fromLabHandout)*

```matlab
% MATLAB Code (Householder to find Tridiagonal)
A = [4 1 2;
     1 3 5;
     2 5 6];

disp('Original Matrix A = ');  disp(A);

% Step 1: Compute S
S = sqrt((A(1,2))^2 + (A(1,3))^2);

% Step 2: Compute vector V = [0 ; x2 ; x3]
x2 = sqrt( (1/2) * (1 + (A(1,2) * sign(A(1,2))) / S) );
x3 = (A(1,3) * sign(A(1,2))) / (2 * S * x2);
V  = [0; x2; x3];

disp('Vector V = ');  disp(V);

% Step 3: Householder matrix P = I - 2*V*V'
I = eye(3);
P = I - 2 * (V * V');

disp('Householder Matrix P = ');  disp(P);

% Step 4: Tridiagonal matrix A1 = P*A*P
A1 = P * A * P;
disp('Tridiagonal Matrix A1 = ');  disp(A1);
```

Listing 1: Base Householder code for a 3×3 matrix

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▶ *Question* Apply the Householder Method to find the tridiagonal form of the following two symmetric matrices and verify the results in MATLAB:

$$A_1 = \begin{bmatrix} 7 & 1 & 4 \\ 1 & 5 & 2 \\ 4 & 2 & 8 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 8 & 3 & 2 \\ 3 & 6 & 1 \\ 2 & 1 & 7 \end{bmatrix}$$

▶ *Theory* For each 3×3 symmetric matrix one Householder step is sufficient ($n - 2 = 1$). The transformation $A_1 = PAP$ preserves the eigenvalues and produces a matrix of the form:

$$A_1 = \begin{bmatrix} * & * & 0 \\ * & * & * \\ 0 & * & * \end{bmatrix}$$

Matrix 1 — key values:

$$S_1 = \sqrt{1^2 + 4^2} = \sqrt{17} \approx 4.1231, \quad x_2 \approx 0.9285, \quad x_3 \approx 0.3714$$

Matrix 2 — key values:

$$S_2 = \sqrt{3^2 + 2^2} = \sqrt{13} \approx 3.6056, \quad x_2 \approx 0.9045, \quad x_3 \approx 0.2774$$

Parameter	Matrix 1	Matrix 2
Size	3×3	3×3
Householder steps	1	1
S	$\sqrt{17} \approx 4.1231$	$\sqrt{13} \approx 3.6056$

Table 1: Settings for Task 01

▶ *MATLABCode*

```
clc;
clear;
close all;

disp('===== LAB 11: HOUSEHOLDER TRIDIAGONALIZATION =====');

%% TASK 1 - MATRIX 1
A1 = [7 1 4;
      1 5 2;
      4 2 8];

disp('Original Matrix 1:');
disp(A1);

T1 = householder_tridiagonal(A1);

disp('Tridiagonal Matrix 1:');
```

```
disp(T1);

%% TASK 1 - MATRIX 2
A2 = [8 3 2;
      3 6 1;
      2 1 7];

disp('Original Matrix 2:');
disp(A2);

T2 = householder_tridiagonal(A2);

disp('Tridiagonal Matrix 2:');
disp(T2);
```

Listing 2: Task 01 — Householder tridiagonalization for two 3×3 matrices

► *Output*

Discussion

Matrix 1 is reduced to tridiagonal form with diagonal entries $[7.0000,\ 8.7647,\ 4.2353]$ and sub-diagonal entries $[-4.1231,\ 1.0588]$. The $(1,3)$ and $(3,1)$ elements are effectively zero (≈ -0.0000), confirming successful tridiagonalization in a single Householder step.

Matrix 2 similarly converges to a tridiagonal form with diagonal $[8.0000,\ 7.2308,\ 5.7692]$ and sub-diagonal $[-3.6056,\ -0.8462]$. Both results verify that a single Householder reflection suffices for 3×3 symmetric matrices, and the transformation $A_1 = PAP$ preserves the matrix eigenvalues since P is orthogonal.

C:\Users\DBZ\Downloads\lab9\untitled2.m

```matlab


disp('===== LAB 11: HOUSEHOLDER TRIDIAGONALIZATION =====');

%% TASK 2 - 4x4 MATRIX
A3 = [4 1 2 3;
      1 5 6 7;
      2 6 8 9;
      3 7 9 10];

disp('Original 4x4 Matrix:');
disp(A3);

T3 = householder_tridiagonal(A3);

disp('Tridiagonal 4x4 Matrix:');
disp(T3);

disp('===== COMPLETED SUCCESSFULLY =====');
```

Command Window

```
>> clear
>> untitled2
===== LAB 11: HOUSEHOLDER TRIDIAGONALIZATION =====
Original 4x4 Matrix:
     4     1     2     3
     1     5     6     7
     2     6     8     9
     3     7     9    10

Tridiagonal 4x4 Matrix:
    4.0000   -3.7417    0.0000    0.0000
   -3.7417   21.5000   -5.3352    0.0000
    0.0000   -5.3352    1.3030    0.2059
    0.0000    0.0000    0.2059    0.1970

===== COMPLETED SUCCESSFULLY =====
>>
```

2 1

▶ *Question* Extend the Householder Method to:

1. A 4×4 symmetric matrix.
2. A general $N \times N$ symmetric matrix (function `householder_tridiagonal`).

Test on:

$$A = \begin{bmatrix} 4 & 1 & 2 & 3 \\ 1 & 5 & 6 & 7 \\ 2 & 6 & 8 & 9 \\ 3 & 7 & 9 & 10 \end{bmatrix}$$

▶ *Theory* For a 4×4 matrix, two Householder steps are required ($n - 2 = 2$).

Step 1 ($k = 1$): Eliminate positions $(1, 3)$ and $(1, 4)$.

$$S_1 = \sqrt{a_{21}^2 + a_{31}^2 + a_{41}^2} = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14} \approx 3.7417$$

$$x_2 \approx 0.8018, \quad x_3 \approx 0.3330, \quad x_4 \approx 0.5000$$

$$\mathbf{V}_1 = [0;\ 0.8018;\ 0.3330;\ 0.5000], \quad P_1 = I - 2V_1V_1^T, \quad A_1 = P_1AP_1$$

After Step 1:

$$A_1 \approx \begin{bmatrix} 4 & -3.741 & 0 & 0 \\ -3.741 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{bmatrix}$$

Step 2 ($k = 2$): Eliminate position $(2, 4)$.

$$S_2 = \sqrt{A_1(3,2)^2 + A_1(4,2)^2}, \quad \mathbf{V}_2 = [0;\ 0;\ x_3;\ x_4]$$

$$P_2 = I - 2V_2V_2^T, \quad A_2 = P_2A_1P_2$$

Final tridiagonal structure:

$$A_2 = \begin{bmatrix} * & * & 0 & 0 \\ * & * & * & 0 \\ 0 & * & * & * \\ 0 & 0 & * & * \end{bmatrix}$$

▶ *MATLABCode*

```matlab
%% TASK 2 - 4x4 MATRIX
A3 = [4 1 2 3;
      1 5 6 7;
      2 6 8 9;
      3 7 9 10];

disp('Original 4x4 Matrix:');
disp(A3);

```

```
T3 = householder_tridiagonal(A3);

disp('Tridiagonal 4x4 Matrix:');
disp(T3);

disp('===== COMPLETED SUCCESSFULLY =====');

% -------- GENERAL N x N FUNCTION --------
function T = householder_tridiagonal(A)
    n = size(A, 1);
    T = A;

    for k = 1 : n-2
        % Extract sub-column below the k-th diagonal
        x = T(k+1:n, k);
        m = length(x);

        % Compute S (norm of x, signed to improve numerical stability)
        S = norm(x) * sign(x(1));

        % Construct Householder vector
        e1      = zeros(m, 1);
        e1(1)   = 1;
        V_sub   = x + S * e1;
        V_sub   = V_sub / norm(V_sub);   % normalize

        % Embed into full n-vector
        V       = zeros(n, 1);
        V(k+1:n) = V_sub;

        % Build Householder reflector and apply similarity transform
        P = eye(n) - 2 * (V * V');
        T = P * T * P;
    end
end
```

Listing 3: Task 02 — 4×4 matrix and general $N \times N$ Householder function

Figure 1: MATLAB output for Task 01 — Householder tridiagonalization of two 3×3 symmetric matrices

- *Output*

- *TridiagonalResult* The 4×4 matrix is reduced to:

$$T \approx \begin{bmatrix} 4.0000 & -3.7417 & 0.0000 & 0.0000 \\ -3.7417 & 21.5000 & -5.3352 & 0.0000 \\ 0.0000 & -5.3352 & 1.3030 & 0.2059 \\ 0.0000 & 0.0000 & 0.2059 & 0.1970 \end{bmatrix}$$

Observation

The general `householder_tridiagonal` function successfully handles both 3×3 and 4×4 symmetric matrices using the same $n - 2$ loop structure. For the 4×4 matrix, **two** Householder steps eliminate all elements outside the tridiagonal band. The resulting matrix has zeros at positions $(1, 3)$, $(1, 4)$, $(2, 4)$ (and their symmetric counterparts), confirming correct tridiagonalization. The sub-diagonal values $[-3.7417, -5.3352, 0.2059]$ and the diagonal $[4.0000, 21.5000, 1.3030, 0.1970]$ are preserved across both tasks, and the output matches the expected MATLAB results exactly. The function is generalized to $N \times N$ input via the loop `for k = 1 : n-2`, making it applicable to any symmetric matrix of arbitrary size.

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This lab successfully demonstrated the implementation of the **Householder Tridiagonalization Method** for symmetric matrices of various sizes in MATLAB. Key findings are summarized below:

- **Task 1** — Two 3×3 symmetric matrices: A single Householder step reduces each matrix to tridiagonal form. Matrix 1 yields diagonal $[7.0000,\ 8.7647,\ 4.2353]$ and sub-diagonal $[-4.1231,\ 1.0588]$. Matrix 2 yields diagonal $[8.0000,\ 7.2308,\ 5.7692]$ and sub-diagonal $[-3.6056,\ -0.8462]$. In both cases the off-tridiagonal elements are zero, confirming the transformation is exact.
- **Task 2** — 4×4 matrix and general $N \times N$ function: Two Householder steps reduce the 4×4 matrix to tridiagonal form with diagonal $[4.0000,\ 21.5000,\ 1.3030,\ 0.1970]$ and sub-diagonal $[-3.7417,\ -5.3352,\ 0.2059]$. The general function `householder_tridiagonal(A)` uses an $n-2$ iteration loop and is applicable to any $N \times N$ symmetric matrix.
- **Key properties verified:** The Householder reflector $P = I - 2VV^T$ is orthogonal and symmetric. The similarity transformation $A_k = P_k A_{k-1} P_k$ preserves eigenvalues, making this method ideal as a preprocessing step for eigenvalue algorithms such as the QR method. For an $N \times N$ matrix, exactly $N-2$ transformations are required.